SAINT MICHAEL COLLEGE OF CARAGA

ARDUINO-BASED AUTOMATED TRASH BIN WITH DEODORIZER USING FSM ALGORITHM

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In Partial Requirements for the Degree
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CERTIFICATE OF RESEARCH APPROVAL

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ABSTRACT

This study designs and implements an Arduino-based automated trash bin with a deodorizer using a Finite State Machine (FSM) algorithm. This study highlighted the problems of waste management, such as poor sanitation and bad smells. It has features such as an ultrasonic sensor for hand/human detection, a servo motor for automated close/open lid, and automated odor detection and Deodorizer. When a hand is detected within range, the FSM algorithm triggers the servo motor for touchless lid access and activates the Deodorizer to eliminate the foul smell. This FSM-based control improves reliability and accuracy, reducing mechanical and user errors. The prototype will be built from readily available materials and suitable for deployment in homes, schools, and workplaces.

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CHAPTER 1

INTRODUCTION

1.1 Project Context

This project features an automatic open/close trash bin with a deodorizer, powered by Arduino, to enhance convenience and hygiene. Without physical contact, the lid operates automatically using ultrasonic sensors and a Finite State Machine (FSM) to coordinate functions. The system turns on the Deodorizer when a preset limit is reached. Unsanitary conditions, germs, and unpleasant odors often result from unseparated waste. Regular bins do not address odor and require manual handling. This invention promotes a safer, cleaner, and healthier environment in offices, schools, and homes.

Waste management has become a significant concern in an increasingly developed and environmentally conscious world. The growing volume of daily waste presents substantial challenges for communities and organizations responsible for responsible and sustainable disposal and recycling [1]. Drawing from previous research, several studies in other countries have implemented smart trash cans with automated lids and real-time monitoring to enhance waste collection efficiency [2]. Most of these projects concentrate on measuring waste levels and accelerating collection, while overlooking odor issues and health risks associated with routine disposal. This distinction highlights the need for solutions that not only automate lid operation but also control or eliminate odors to improve hygiene.

The study at Divine Word College of San Jose in the Philippines suggested adjusting the sensitivity of Passive Infrared (PIR) sensors (which detect motion based on heat) to reduce false alarms in smart bins [3]. Our project includes a servo motor (a device for precise control of movement) with an ultrasonic sensor to automate the opening of the lid. The ultrasonic sensor measures distance, and when a hand is sensed within its range, the Arduino commands the servo to open. Other materials include a gas sensor (for detecting odors), a deodorizer (consisting of a relay, an electrically operated switch, plus a fan/pump), buzzer or LED (for alert/visual feedback), and optional audio prompts, all coordinated by FSM-based logic for optimized functionality.

This discovery is crucial as it offers a practical solution to frequent unpleasant smells from household trash [4]. While other projects make bins more accessible and reduce contact, they lack deodorizing systems or efficiency in using FSM algorithms for bin operations. There is a gap in creating an automated, deodorizing, hygienic, and user-friendly bin suited to community needs [5]. Ongoing virus spread continues due to surfaces sheltering pathogens, presenting a significant risk of transmission through indirect contact. This research focuses on controlling such issues by developing a Self-Disposing Contactless Motion-Activated Trash Bin with ultrasonic sensors for precise detection [6].A significant challenge in waste management today is the lack of cost-effective automated solutions that communities can readily adopt. Additionally, many people lack motivation to recycle due to the absence of direct incentives. Existing waste disposal methods depend mainly on manual labor, while inadequate integration of technology and community involvement further diminishes the effectiveness of current recycling programs [7].

This study aims to benefit several groups: (a) households and students by reducing direct contact with bin lids and improving indoor comfort through better odor control; (b) waste handlers by lowering early exposure to foul smells and harmful substances; and (c) research and education communities by providing a tested prototype driven by FSM and usability data that can help guide larger implementations. In summary, a low-cost, automated bin with proven deodorizing features can enhance daily sanitation, reduce exposure to bacteria, and serve as a pilot technology for broader adoption in institutions.

1.2 Literature Review

This chapter discusses the related literature on Arduino, specifically the Automated Trash bin, Deodorizer, and the FSM algorithm, to develop and promote a safer, cleaner, healthier atmosphere in schools and households. To achieve this, it presents the flow or the process of an Arduino-based automated Trash Bin with a Deodorizer Using the FSM Algorithm. In addition, it explains the accuracy, limitations, and benefits of optimizing waste management and maintaining a cleaner living space.

Arduino Uno Microcontroller

Recent studies using Arduino boards have greatly improved how well students learn by doing, particularly in coding. Students find them easy to use and a good way to stay interested in learning about microcontroller circuits. University students showed clear improvements in their test scores after using Arduino, especially in coding and having a positive view of the technology [8], [9]. How the platform looks and works makes connecting what you learn about programming to building things easier. These help people with different experience levels learn better [10], [11]. Recent studies have compared Arduino to other similar platforms. One study in 2021 looked at the Arduino Uno and Raspberry Pi Pico [12]. It found that Arduino works well and is suitable for teaching and creating initial versions of projects. However, newer options might be better for tasks that need more processing power. Other studies have examined how Arduino can be improved for complex functions like artificial intelligence. These studies confirm that Arduino is easy to use, reliable, and can be used in many different situations [13], [14]. Research from 2021 to 2025 shows that Arduino boards are still functional, easy to get, and adaptable for learning, creating new technologies, automating lab work, and building real-world applications. New studies highlight how easy they are to use,

how well they can be adjusted, and how they fit into complex work processes. These ensure Arduino remains vital in both schools and industries [15], [16], [17], [18].

Automated Trash Bin Powered by an Arduino

Arduino-powered trash cans are becoming a popular way to make waste management more efficient, cleaner, and easier for people to use. Recent studies have different designs, from simple automated open/close lids to more advanced systems that connect to the internet and sort waste. These bins use Arduino boards (like Uno or Nano) as the central control system. Their Bin uses sensors to detect when someone is near the Bin or when the Bin is complete, and motors to automatically open and close the lid [19], [20], [21], [22], [23]. Some bins have added features like self-moving robots to transport them, mechanisms that compress the trash to hold more, and even built-in hand sanitizer dispensers for better hygiene [3], [24], [25], [26]. Real-world evaluation shows that trash bins powered by Arduino are usually accurate, efficient, and easy to use. They successfully automate the lid opening and accurately detect waste levels [19], [23]. These systems help promote cleanliness by reducing the need for people to touch the Bin, which is good for public health and makes them more accessible, especially for people with disabilities [3]. Math models have also been used to evaluate the system's reliability and identify the most critical maintenance components [27]. Arduino-based trash bins are enhancing waste management significantly. They let you throw trash without touching the lid manually, another feature is to track how full the Bin is, and even sort the garbage. These systems are reliable, easy to grow, and fit into any smart city. They handle trash in a cleaner, more efficient, and hygienic way [24].

Finite State Machine Algorithm

Finite State Machine (FSM) algorithms are fundamental in computer science and engineering because they offer an organized method for simulating sequential logic, control systems, language processing, and other topics. Recent research has broadened their theoretical support, real-world applications, and integration with cutting-edge technologies. Additionally, robust control techniques have been developed to maintain desired behaviors under uncertainty. These techniques are found in embedded systems, communication networks, and industrial systems [28]. Testing and verifying systems based on Finite State Machines (FSM) remains significant. Recently, algorithms that utilize reinforcement learning artificial intelligence capabilities have been proposed to speed up FSM-based testing, reducing both the computational cost and memory overhead in generating distinguishing and reset sequences for advanced systems [29]. Finite State Machine Algorithm studies have developed algebraic analysis, robust control, hardware optimization, and the integration of finite state machines with neural networks. These advances have all contributed to the finite state machine's efficiency, reliability, and applicability across a growing range of contemporary computational and engineering systems [28], [30].

Accuracy and Reliability of the FSM Algorithm

Recent studies have evaluated the application of finite state machine (FSM) algorithms on microcontroller devices, including Arduino, and their ability, reliability, and efficiency while working in real-world embedded and IoT environments. A study in 2022 developed a hybrid algorithm that combined an (RSVM) with an FSM to detect gait phases in real time on wearable microcontroller devices [31]. The FSM algorithm confirmed and refined the RSVM predictions, significantly enhancing the system's reliability. The research implemented the

gait detection algorithm on a microcontroller, where high real-time performance was achieved, at an accuracy of 91.51%, sensitivity of 91.70% and specificity of 95.77%, despite changing real-world conditions. Again, this demonstrated the FSM's valid decision-making capacity in terms of being robust and relying on a hardware configuration similar to Arduino. Research shows that FSM algorithms are effectively applied to Arduino-based systems, particularly for structuring communication protocols and enabling adaptive behaviors in IoT networks. Ongoing theoretical advances further support the reliability and scalability of FSM implementations in embedded platforms like Arduino [30], [32], [33], [34].

Deodorizer System Powered by an Arduino

A study was conducted in which metal oxide semiconductor sensors were installed on an Arduino to identify human scent patterns. Analog signals from the sensors were converted to digital signals by the Arduino and then analyzed with Artificial Neural Networks (ANNs) for feature extraction and classification. Overall, the results demonstrated the system could identify odors with 85% accuracy, indicating Arduino's potential for predictable and real-time odor monitoring and pattern recognition, which is desirable in an innovative deodorizing system. Recent studies have also examined Arduino-based air quality monitoring systems [35]. For example, a study in 2022 created a long-range IoT air quality monitoring system that used an Arduino and various gas sensors, which measured NO2, SO2, CO2, temperature, and humidity. The air quality monitoring system effectively gathered and streamed air quality data in real time, which shows the Arduino is a reliable platform for continuous monitoring of the environment, while also allowing for a connection to deodorizing devices [36]. These Studies confirm that Arduino-based systems are highly

effective for odor detection and air quality monitoring, providing a strong technological foundation for developing automated deodorizer solutions [37].

1.3 Objective of the Study

The researchers aim to design and build an Arduino-based automated trash bin equipped with a deodorizer. It will be controlled/programmed through a Finite State Machine (FSM) algorithm. This project aims to improve cleanliness, hygiene, and convenience in waste disposal at offices, classrooms, and faculty areas.

Specifically, it aims to:

1. Design a contactless trash bin with an ultrasonic sensor that automatically opens and closes the lid. These will reduce physical contact in offices, classrooms, and faculty rooms.
2. Integrate a deodorizer system that activates only when trash is disposed of and odors are detected to minimize unpleasant smells in enclosed areas.
3. Implement a Finite State Machine (FSM) algorithm to control the lid movement and deodorizer operation effectively.
4. Assess the system's functionality and accuracy in maintaining cleanliness and user convenience using measurable metrics such as: Sensor Detection Accuracy, Lid Response Time, FSM State Accuracy, Deodorizer Activation Accuracy, Odor Reduction, User Satisfaction Score, and Contact Reduction.

1.4 Scope and Limitations

The study aims to use the Finite State Machine (FSM) algorithm to design and develop an automated trash bin with a deodorizer based on an Arduino. This project aims to make waste disposal easier and cleaner. The automated trash bin will be used in faculty areas, offices, and classrooms where odor control and cleanliness are essential. The main functions include a deodorizer that reduces unpleasant smells and a contactless lid that operates through MQ-135 and ultrasonic sensors. The system aims to give faculty, staff, and students a practical solution. The FSM algorithm systematically manages the process. It ensures correct operation and energy efficiency by detecting waste foul smells, opening and closing the lid, and turning on the Deodorizer when necessary.

Despite its innovative features, the system has limitations. The prototype is intended for indoor use and is only made to fit small and medium-sized trash cans. These restrict its capacity to manage outdoor or large-scale waste. The Deodorizer may not eliminate strong or chemical odors; it only works with simple spray systems. Accurate sensor detection and a steady power source are essential to the system, and external elements like obstructions or lighting can impact both. Moreover, the project does not include sophisticated waste sorting, recycling techniques, and extensive disposal management. Its use in other institutions or broader contexts is limited because it is primarily designed for offices and classrooms.

1.5. Definition of Terms

The terms below explain how they are used in this study and describe the core principles and components in developing the Arduino-Based Automated Trash Bin with Deodorizer Using FSM Algorithm.

Arduino Uno/ Microcontroller Development Board—The trash bin's main microcontroller. It processes the sensor data, executes the FSM algorithm, and controls the lid and deodorizer system.

Automated Trash Bin – The prototype designed in this study provides contactless lid operation and odor management features. It is intended for use in classrooms, offices, and faculty rooms.

Deodorizer System – A spray mechanism integrated into the Bin activates only when odors are detected or trash is disposed of. It minimizes unpleasant smells in enclosed school areas.

Finite State Machine (FSM) Algorithm – A programming model used in this study to control the sequence of operations, including lid movement and deodorizer activation. It ensures the trash bin's systematic, efficient, and reliable operation.

MQ-135 Sensor – A gas sensor used in this study to detect air quality changes inside the trash bin, such as unpleasant odors and harmful gases. It triggers the Deodorizer when odors are exceeded.

Ultrasonic Sensor – Detects the presence of hands or objects near the trash bin. It enables contactless lid operation by signaling the Arduino to open or close the lid.

Relay Module – An electronic switch used in the prototype to control the deodorizer system, turning it on or off based on sensor inputs.

Sanitation - refers to improving cleanliness and hygiene within facilities, which was achieved by implementing an automated trash bin with a deodorizer.

Servo Motor – A motor responsible for controlling the trash bin lid's movement, ensuring smooth opening and closing operations.

User Convenience – The increased comfort and efficiency provided to faculty, staff, and students through reduced physical contact with bin lids and improved odor control.

CHAPTER 2

TECHNICAL BACKGROUND

Research Design

This study employs an Experimental Developmental Research Design to create an automated trash bin with a deodorizer. The process involved identifying common waste disposal issues, designing the Bin's structure using Arduino and sensors, and integrating the Finite State Machine (FSM) algorithm for system control. After development, the prototype was tested to evaluate its performance regarding automatic lid operation, odor-eliminating system, and overall reliability. These allowed for refinements and improvements.

2.1 System Architecture

This chapter presents a detailed analysis of the **ARDUINO-BASED AUTOMATED TRASH BIN WITH DEODORIZER USING THE FSM ALGORITHM**. It describes the devices, components, and control logic required to install and operate the system. The distinctive characteristics of the system are also discussed, including the integration of the Finite State Machine (FSM) algorithm, which effectively governs and controls the automated functions of the trash bin—such as lid operation and deodorizer activation.

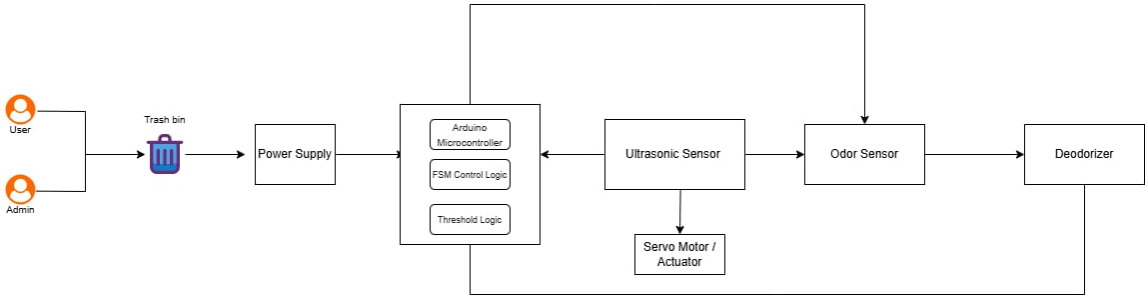


Figure 1. System Architecture

Figure 1 shows the system architecture, which shows how the automated trash bin uses Arduino functions. The users will be involved in the trash bin, which has been provided with a stable power supply that supplies power to all system components. The Arduino microcontroller and the Finite State Machine (FSM) control logic form the primary processing unit capable of handling all input and output functions. These were done through the ultrasonic sensor that reads when a person's hand is close to the Bin, and the servo motor will automatically open or close the lid. At the same time, the odor sensor measures the level of smell in the Bin, and once foul odors are detected, the lid will close, and the Deodorizer is turned on to remove the bad smell. Altogether, the system is effective due to the arrangement of sensors, actuators, and control logic that guarantee an automatic, hygienic, and energy-efficient waste disposal process.

2.2 Conceptual Diagram

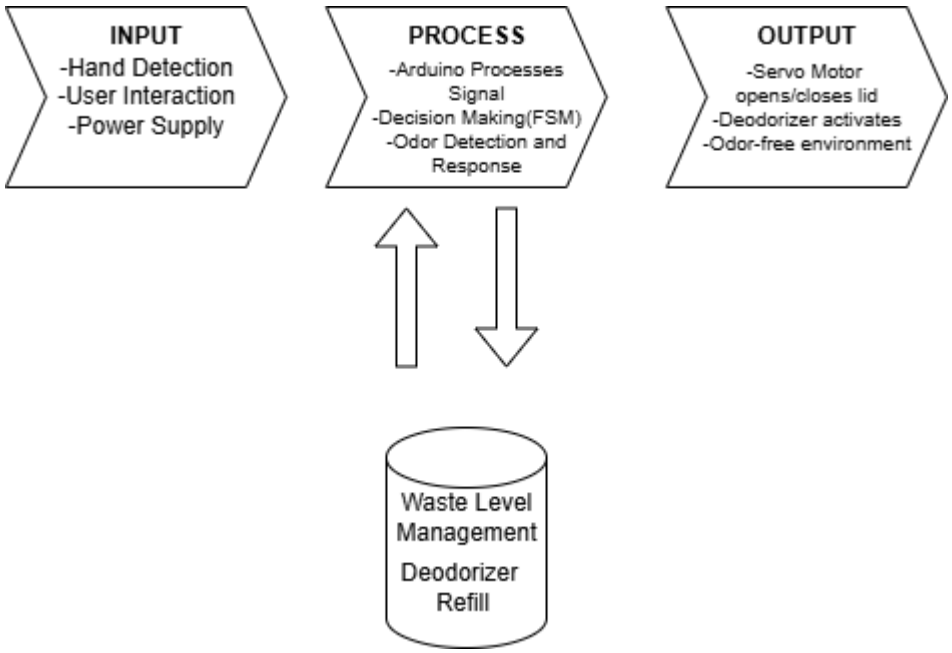


Figure 2. Conceptual Diagram

Figure 2: The Conceptual Diagram shows the general flow of the prototype and the functional relationship in the Arduino-Based Automated Trash Bin with Deodorizer Using FSM Algorithm. The user is the first node of the system to activate the process with the help of the hand detection system and the lid control system. These are the inputs to the system; the trash bin can detect the presence of the hand and automatically open and close its lid. The process stage has several internal processes that the system components must handle. The Arduino board includes Finite State Machine (FSM) logic that manages all the automated processes, including power, lid movement, and odor detection. The power management ensures power is used efficiently, and the deodorizer mechanism is triggered to eliminate unpleasant odors in the Bin. The system also closes its lid first before activating the deodorizer mechanism. The Diagram shows that the input signals are processed and converted into practical and automated waste management activities.

2.3 Use Case Diagram

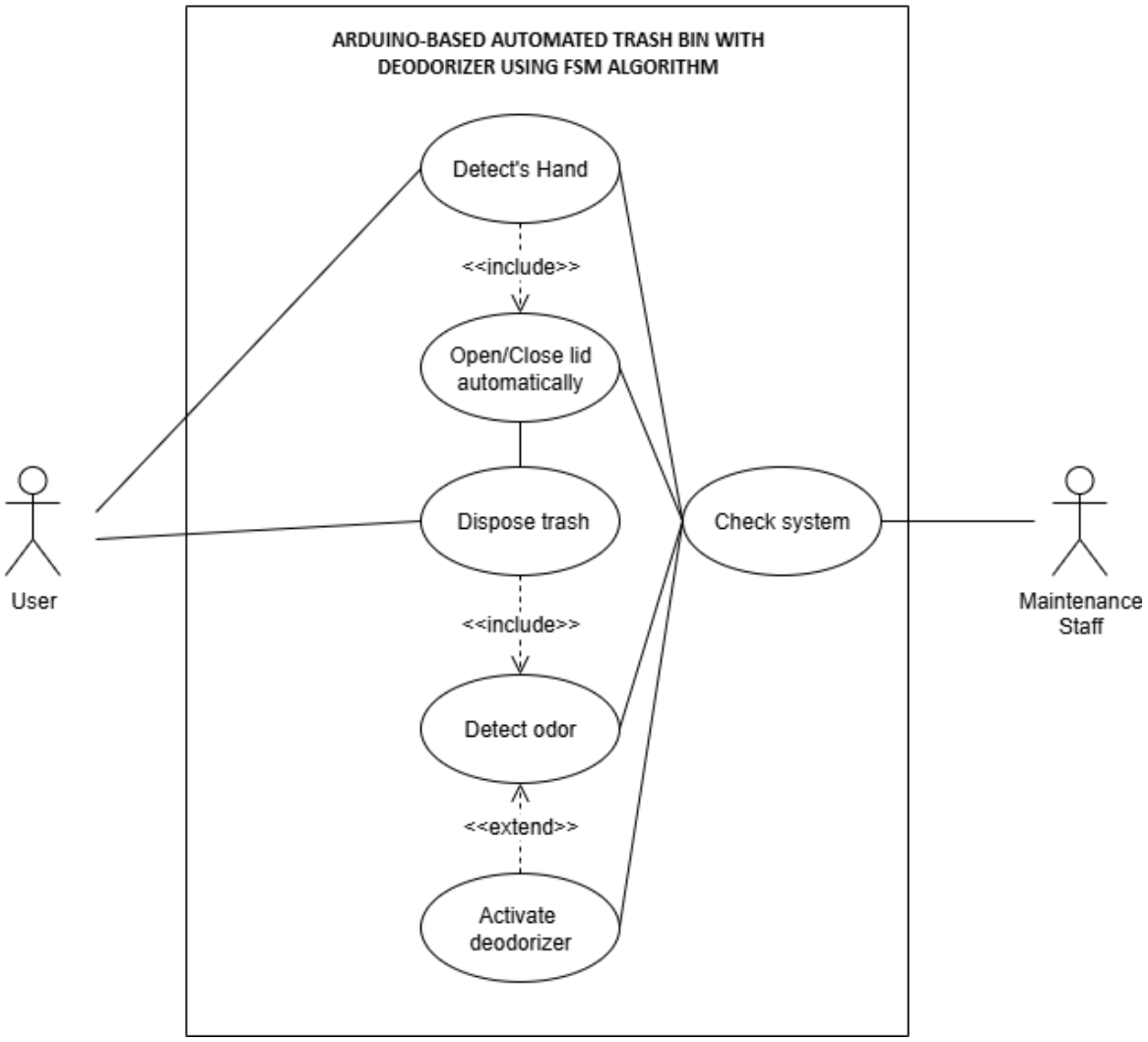


Figure 3. Use Case Diagram

Figure 3: The system automatically detects a hand placed near the trash bin using sensors. When an item is sensed, the device triggers a way to lift a lid automatically so that you can throw trash away without touching anything. The lid closes automatically after the hand is no longer detected. Keeping your hands clean and your room free from germs and odors. The detector also activates the deodorizer dispenser, which discharges a nice-smelling scent to counteract any foul odor from the waste. This feature makes it easier and more sanitary to dispose of trash.

2.4 Activity Diagram

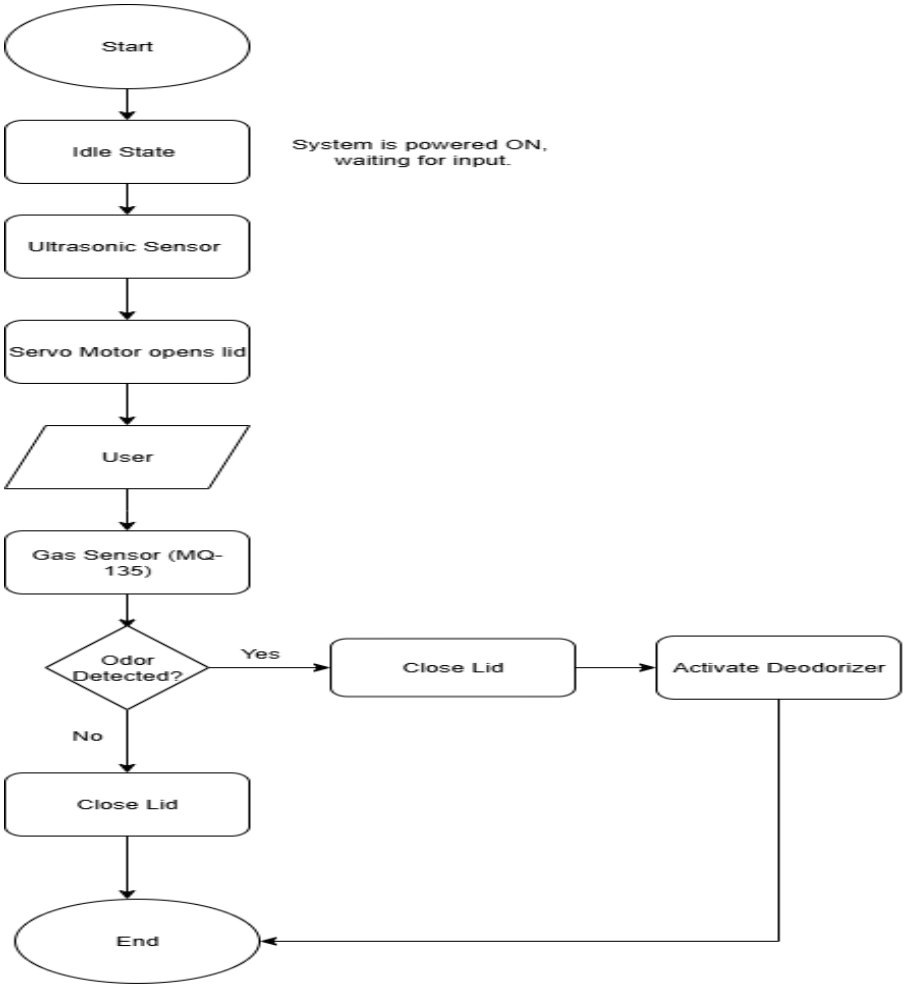


Figure 4. Activity Diagram

Figure 4. The cycle starts with the system being switched on, and it switches to an idle state, where it is not active but constantly surveys its surroundings and waits to be touched. The ultrasonic sensor in this state detects the presence of a user in front of the trash bin. When the sensor detects that there has been movement within a given range, it will signal the microcontroller, which in turn will send a signal to the servo motor to open the lid automatically. These will enable the user to discard waste without physical contact, enhancing hygiene and convenience. Once the waste is deposited, the system activates a gas sensor (MQ-135) to check the internal conditions of the Bin, whether it has a foul odor or not. The gas sensor analyzes air

quality by identifying forbidden or unpleasant gases that could be released due to garbage decomposing. If the sensor concludes that there is no odor of significance, the system will close its lid and move back to the idle position, awaiting the next user interaction. Once the waste is deposited, the system activates a gas sensor (MQ-135) to check the internal conditions of the Bin, whether it has a foul odor or not. The gas sensor analyzes air quality by identifying forbidden or unpleasant gases that could be released due to garbage decomposing. If the sensor concludes that there is no odor of significance, the system will close its lid and move back to the idle position, awaiting the next user interaction.

2.5 Sequence Diagram

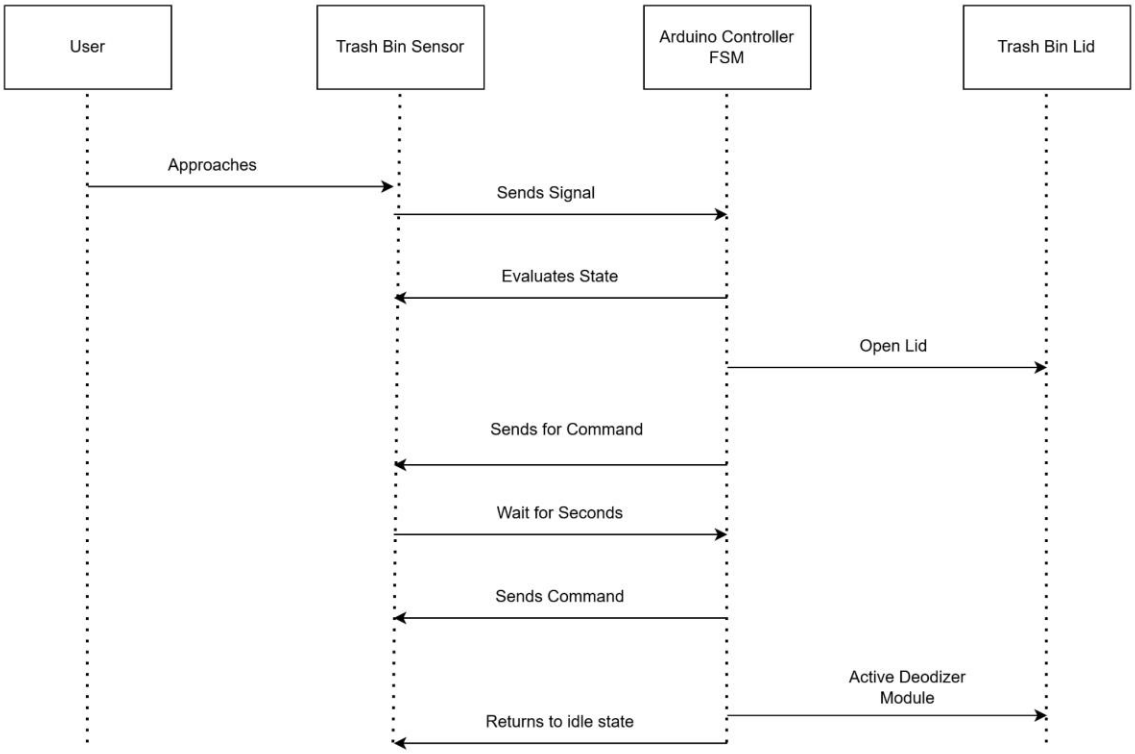


Figure 5. Sequence Diagram

Figure 5 presents the detailed operation of the automated trash bin system driven by an Arduino with the help of a Finite State Machine (FSM) algorithm. The sequence starts when the user goes to the Bin and the ultrasonic sensor detects an object nearby, sending a signal to the FSM controller. The FSM controller combines the signal and switches the system into the Open Lid state by giving a command to the motor system. When the lid is opened, the system takes a few seconds to enable the user to clear away the waste. Then the FSM controller orders the motor to shut the lid, and simultaneously, it orders the deodorizer module to produce a fresh smell, eliminating bad smells. Once all activities are through, the FSM controller returns to the Idle state, gearing the system to the next user interaction. In general, this sequence diagram shows the

coordinated flow of communication between the user and the sensors, controller, and the output modules to have a fluid, hygienic, and automated waste disposal cycle.

2.6. State Diagram (Algorithm)

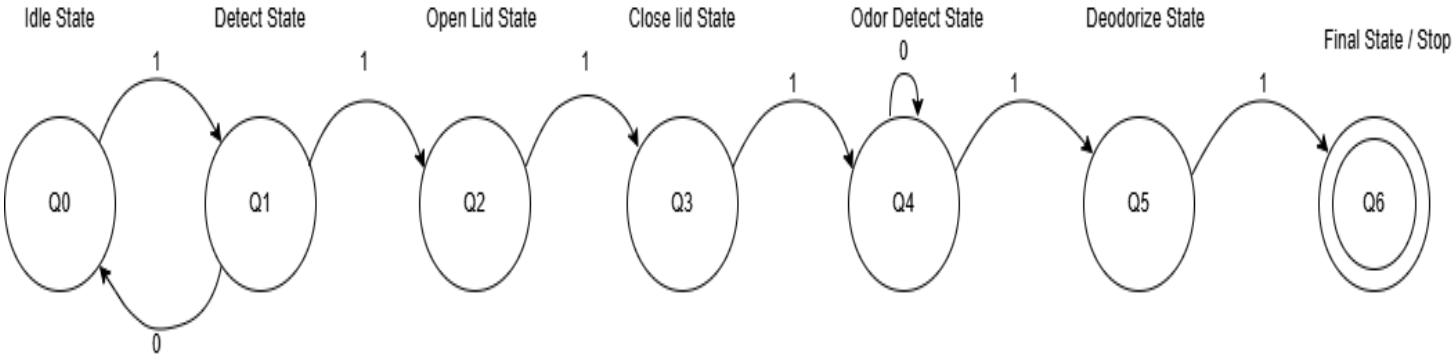


Figure 6. State Diagram

Figure 6: The queue starts in the Idle State (Q0), waiting for input. When an object is detected (Detect State Q1) below a sensor (input '1'), the transition to detect state is made; when no objects are detected (input '0'), the system returns to Idle State to save energy. Subsequently, the lid opens (Open Lid State Q2) with an input of '1' to dispose of waste, and finally closes (Close Lid State Q3) again with an input of '1', preventing bad smell from escaping outside. The system now goes to the Q4 state (Odor Detect State), continuously checking for odor. If so, it inputs '0' and goes into the deodorize state (Q5) with input '1', starting deodorization. If the output is '1', then the system has arrived at its last state/Final State/Stop (Q6) and stops or completes its cycle, after which it again waits for a detection. This process provides automatic sanitary waste disposal and odor control.

2.7. User Interface Design (Prototype)

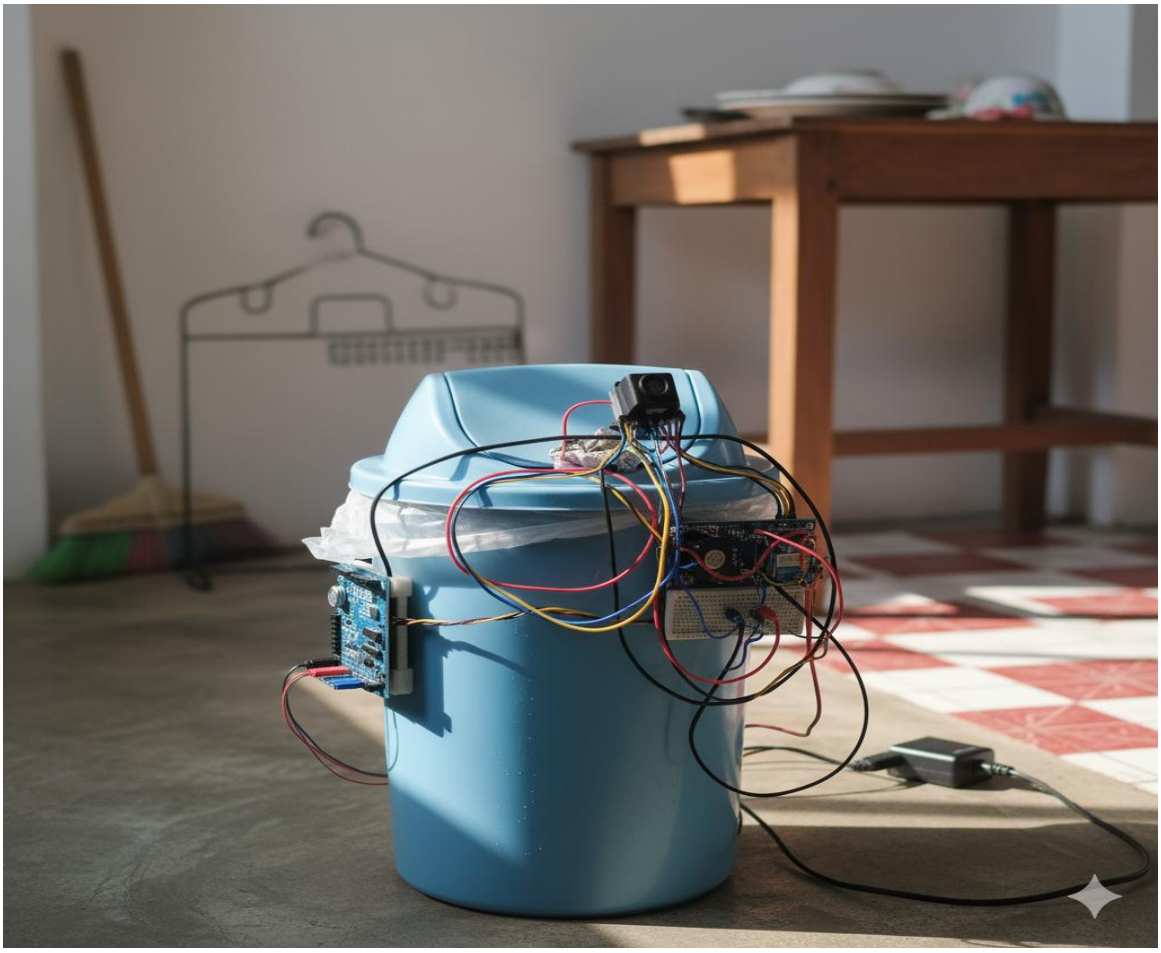


Figure 7.1 Trash Bin Prototype

Figure 7: A user-friendly and easy-to-understand interface to visually depict the current state of the Bin (idle, detecting, open/close lid, or deodorizing), show sensor data, and controls most actions, such as the opening of the lid or the Deodorizer, and visually reflects the change of state based on sensor data, which guarantees effective monitoring, interaction, and system feedback.

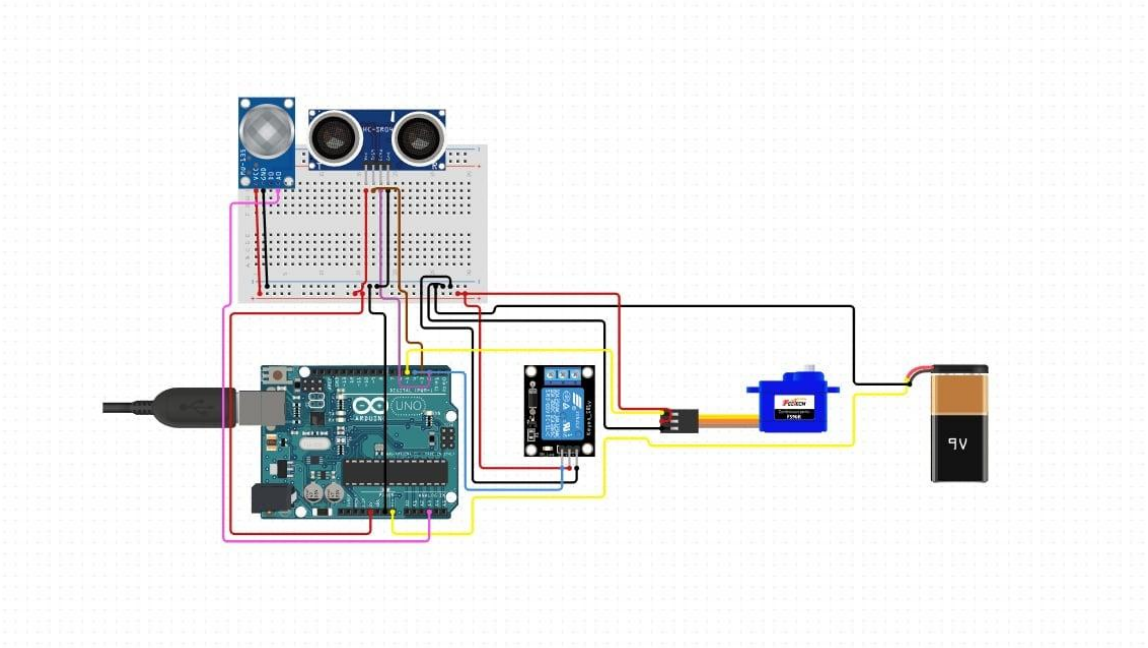


Figure 7.2 Wiring Diagram / Circuit Design

Figure 7: The system contains an ultrasonic sensor, a servo motor, a relay module, a deodorizer, and a 9V power source. The ultrasonic sensor has been fitted on the top of the trash bin to sense the distance of an approaching object or hand. The sensor sends a signal to the Arduino board upon sensing an object within a predefined range. The Arduino Uno processes this signal, and the servo motor is commanded to rotate, and the trash bin lid opens automatically. The servo motor takes a little time to return to its initial position, and the lid closes. The Arduino also controls the relay module (connected to the Deodorizer). After closing the lid, the Deodorizer goes on and emits a small puff of fragrance that will assist in getting rid of unpleasant smells present in the Bin. The circuit power supply consists of a 9V battery, the primary power supplier to the servo motor and the Deodorizer.

In contrast, the Arduino board is powered through USB or an external adapter. The breadboard is the focal connection point of all the components that provide stability of electrical connections and efficient transmission of signals between devices. This design illustrates how the system

incorporates sensing, actuation, and control using the finite state machine (FSM) algorithm to automate the system and ensure hygiene efficiency fully.

2.7 Software Platforms, Development Environment and Tools Software Platforms

Table 1

Development Environment

Components	**Specification**	**Usage**
Programming Language	C/C++ (Arduino IDE)	Used to program the Arduino microcontroller for controlling sensors, motors, and logic.
Libraries	Servo.h	Controls the servo motor for opening and closing the bin's lid.
	MQ135.h (if used)	Reads gas concentration or odor levels to detect unpleasant smells.
	NewPing.h (optional)	Simplifies ultrasonic sensor readings and enhances distance measurement accuracy.
FSM (Finite State Machine)	switch-case statements	Provides structured decision-making for bin operations, ensuring reliable state transitions.

Table 2

Tools

Components	Specification	Usage
USB Cable	Standard USB Type A to B	Used to upload code from the Arduino IDE to the Arduino board.
Serial Monitor	Built-in Arduino IDE tool	Tests and debugs the Arduino code by displaying real-time sensor values and system messages.
Multimeter	Digital or Analog	Measures voltage, continuity, and checks electrical connections for troubleshooting.
Breadboard & Wires	Standard prototyping tools	Allows temporary circuit building and testing without soldering.
Soldering Kit	Soldering iron, lead, flux	Used for permanent circuit assembly and stronger hardware connections.
Power Supply	Battery (9V) / Adapter	It provides the necessary power to the Arduino and connected components.

2.8 Hardware Requirements

Table 3

Hardware Requirements

Components	Specification	Usage
Control Unit	Standard USB Type	Used to upload code from the Arduino IDE to the Arduino board.
Sensors	MQ-135 Sensor	It detects the presence of several Gases by varying their resistance; It applies to air quality and odor monitoring.
	Ultrasonic Sensor (HC-SR04)	It measures distance using ultrasonic waves, which are typically used in automation and obstacle detection.
Actuators	Relay Module	Operates electronic circuits with higher power consumption (e.g., pumps, fans) using a low-power control signal.
	Servo Motor (SG90)	It provides accurate angular motion when driven by electrical current and is used to open/close the trash bin lid.
Indicators	Buzzer / LED Indicators	Buzzer provides audio alerts or notifications; the LED emits light to indicate system status.
Power	Power Supply	Provides a consistent supply of electrical power (via batteries or adapter) to the microcontroller and components.
Prototyping Tools	Jumper Wires	Flexible cables are used to connect electronic components.
	Breadboard	A reusable circuit board for testing circuits without soldering.
	PCB	Permanently joins components Using copper tracks for a finalized hardware design.
Passive Components	Resistors	Restrict or regulate the flow of electric current within the circuit.

Mechanical	Trash Bin	The physical container that houses the automated system and electronic parts.

2.9 ETHICAL STANDARD

This study adhered to the ethical guidelines of Saint Michael College of Caraga and followed the standards set by the Research Ethics Committee. All procedures involving human participants were conducted with respect for their rights, safety, privacy, and voluntary participation. The project ensured responsible data handling, transparent reporting, and strict compliance with Philippine laws on intellectual property and data privacy. No animal testing was conducted, and all human involvement was limited to non-invasive observation and feedback. The researchers exercised honesty, responsibility, and accountability throughout the development, testing, and documentation of the system.

A. Protection of Intellectual Property Rights (IPR)

The "Arduino-Based Automated Trash Bin with Deodorizer Using FSM Algorithm" is an original work developed by the research team. All design components, hardware layout, documentation, and FSM logic are original works protected under Republic Act No. 8293 (Intellectual Property Code of the Philippines) as copyright. The system does not use any unique logo or trademark. External materials—such as Arduino open-source libraries, diagrams, and reference literature—were cited correctly and attributed. The researchers complied with open-source licenses (e.g., the MIT/GPL licenses). Reproduction, distribution, or modification of the project without written permission from the authors is strictly prohibited.

B. Informed Consent

Written informed consent was **distributed** to all participants before prototype evaluation. The form explained the study's purpose, procedures, expected benefits, minimal risks, and data to be collected. Participants were assured of their **right to withdraw at any time without penalty**, and the consent form used simple, accessible language to ensure understanding. No animal trials or invasive procedures were involved. Only individuals who voluntarily agreed and signed the consent form took part in the evaluation.

C. Data Privacy and Confidentiality

All collected data—including sensor performance results and participant feedback—were treated with **strict confidentiality** in compliance with **Republic Act No. 10173 (Data Privacy Act of 2012)**. Digital files were stored in password-protected devices with restricted access, while physical documents were kept in secured storage. Data were anonymized before analysis, and no personally identifiable information was included in reporting. Anonymized data may be used for academic purposes upon request. All data will be retained only for the duration required by the institution and will be destroyed afterward through secure deletion or shredding.

D. Voluntary Participation and Freedom to Withdraw

Participation was entirely voluntary, with no coercion, incentives, or pressure applied. Respondents were informed of their freedom to decline or discontinue their participation at any point without consequences. This principle was emphasized in both the orientation and the written consent form.

E. Minimization of Harm and Risk Management

The prototype was handled in controlled, safe conditions, following institutional safety protocols for electrical wiring and chemical handling. All electronic components were properly insulated to prevent electrical hazards. The deodorizer used was **non-toxic**, safe for users, and compliant with environmental safety guidelines. No physical, psychological, or emotional harm occurred during testing. A structured **risk assessment** was performed before each trial to ensure participant safety and environmental protection.

F. Beneficence and Contribution to Knowledge

The project provides practical community benefits by promoting hygienic, contactless waste disposal and reducing odor through an affordable, environmentally friendly system. Findings contribute to academic knowledge in automation, FSM applications, and sustainable technology. Research results may be shared for academic reference upon request, supporting further studies in environmental and automation innovation.

G. Justice and Fair Participant Selection

Participants were selected based on availability, relevance, and willingness to participate. Inclusion criteria focused solely on their ability to provide meaningful feedback on system functionality. No exclusions were made based on gender, age, religion, or socioeconomic background. Selection was equitable and free from any form of discrimination.

H. Data Integrity and Accuracy

All sensor readings, response measurements, and deodorizer performance results were accurately recorded using calibrated tools and standardized procedures. Multiple trials were conducted to ensure reliability and minimize bias. No data manipulation or selective reporting occurred. AI tools used during documentation did not influence technical data, analysis, or results.

I. Transparency and Honesty in Reporting

All findings, challenges, and limitations of the study were **reported** truthfully. No falsification, fabrication, or manipulation of data occurred. External materials were cited correctly, and the researchers declared that they had no conflicts of interest that would affect the study's results.

J. Use of Patented or Copyrighted Materials

The project used standard Arduino components and open-source libraries under legally permissible licenses (MIT/GPL). Copyrighted diagrams, texts, or references were cited correctly. Patent restrictions did not cover the deodorizer mixture used. No patented materials requiring licensing or royalties were utilized.

K. Ethical Considerations for Animal and Human Trials

The study did not involve animal testing or invasive human procedures. Human involvement was limited to voluntary observation and evaluation of prototype operation. All activities complied with ethical guidelines for non-invasive human research, ensuring participant safety and well-being.

L. Responsible Use of AI and Related Technologies

AI tools such as ChatGPT, Grammarly, and QuillBot were used **only to improve the clarity, structure, and presentation of documentation, not to generate data, conduct analysis, or produce** technical outputs. The researchers verified all information processed by AI. Transparency regarding AI assistance is fully disclosed in the study.

M. Ethical Clearance and Institutional Approval

The study was **reviewed and approved by the Research Ethics Committee of Saint Michael College of Caraga** before data collection and prototype testing. Approval confirms

that the research complied with all institutional, ethical, and safety requirements. (Date of approval may be added if required.

Data Gathering Procedure

The data gathering process was carried out in five systematic stages:

1. **Prototype Testing**: The automated trash bin prototype was tested repeatedly under controlled and varied environmental conditions to ensure consistency and validity of results. Each test cycle was performed at least 10 times per condition to measure the ultrasonic sensor detection accuracy, lid response time (in seconds), and the efficiency of the deodorizer system. Timers, distance markers, and observation checklists were used to systematically record all measurements.

2. **Collection of User Feedback**: A group of volunteer participants was provided with the prototype and asked to operate it following a standardized set of instructions. Before participation, written informed consent was obtained, and all responses were recorded anonymously. User impressions of usability, cleanliness, odor control, and convenience were collected through a structured questionnaire and an evaluation checklist, ensuring uniformity in data collection.

3. **Performance Evaluation**: The system's operational performance was assessed through quantifiable indicators, including the number of successful lid activations, deodorizer activation rate, and power consumption (in milliamperes or watts). Measurements were documented using digital logs, observation sheets, and calibrated measuring tools, following repeated trials to establish reliability and consistency.

4. **Data Handling**: Both statistical (e.g., mean, percentage accuracy, and response-time computation) and qualitative methods (e.g., thematic coding of user feedback) were applied to analyze the collected data. Processing and tabulation were conducted in Microsoft Excel, with confidentiality maintained, restricted data access, and secure file storage throughout the analysis phase.

5. **Reporting and Dissemination**: The validated findings were formally presented to the research adviser, panel members, and stakeholders in accordance with institutional research standards. All results were reported in a clear, accurate, and confidential manner, ensuring compliance with ethical guidelines and maintaining the privacy of all participants.

CHAPTER 3

SOFTWARE DEVELOPMENT AND TESTING

4.1 Development Process

4.2 Testing Process

- Accuracy Test

-Functionality Test

-Performance Test

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Membership/Affiliations

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APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo
B - Instrument with Informed Consent
C - Map of the Research Locale *(Include Coordinates)*
D - Curriculum Vitae
E - Narrative & Photo Documentation
F- Ethics Committee Approval

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